Algebra & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Name, identify, order, and compare fractions**

Rationalise the denominator and simplify: $\frac{10}{3-\sqrt{5}}$.

Multiply numerator and denominator by the conjugate $3+\sqrt{5}$:
$\frac{10(3+\sqrt{5})}{(3-\sqrt{5})(3+\sqrt{5})} = \frac{10(3+\sqrt{5})}{9-5} = \frac{10(3+\sqrt{5})}{4} = \frac{5(3+\sqrt{5})}{2} = \frac{15+5\sqrt{5}}{2}$. [M1 for multiplying by conjugate, M1 for difference of squares denominator, A1 for simplified form]

2 **[Foundational]** • **Target: Equivalent and simplified fractions**

A bag contains 6 green counters and 4 yellow counters. A counter is chosen at random, its colour recorded, and it is REPLACED. A second counter is then chosen. What is the probability that both counters are the same colour?

Since the counter is replaced, events are independent: $P(\text{Green, Green}) = \left(\frac{6}{10}\right)^2 = \frac{36}{100}$
$P(\text{Yellow, Yellow}) = \left(\frac{4}{10}\right)^2 = \frac{16}{100}$ $P(\text{Same Colour}) = \frac{36}{100} + \frac{16}{100} = \frac{52}{100} = \frac{13}{25}$.

3 **[Foundational]** • **Target: Adding and subtracting fractions**

A car travels a distance of 150 km, correct to the nearest 10 km, in a time of 2 hours, correct to the nearest 0.1 hour. What is the upper bound of the average speed in km/h?

Upper bound of speed = $\frac{\text{Upper bound of distance}}{\text{Lower bound of time}}$. Upper distance = $150 + 5 = 155$ km. Lower time = $2.0 - 0.05 = 1.95$ hours. Maximum speed = $\frac{155}{1.95} \approx 79.487$ km/h ≈ 79.5 km/h. [M1 for upper dist / lower time, A1 for 79.5]

4 **[Foundational]** • **Target: Fractions of a quantity**

In a survey of 50 tourists, 30 visited museum M, 25 visited palace P, and 20 visited art gallery G. 12 visited both M and P, 10 visited both P and G, 8 visited both M and G, and 5 visited all three. How many tourists visited NONE of the three attractions?

By the 3-set inclusion-exclusion principle: $n(M \cup P \cup G) = (30+25+20) - (12+10+8) + 5 = 75 - 30 + 5 = 50$. Since the union equals 50, all 50 tourists visited at least one attraction. Therefore, $50 - 50 = 0$ tourists visited none. [M1 for 3-set formula, A1 for 50, A1 for 0]

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5 [Foundational] • Target: Multiplying fractions

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of USD 2000 to exceed USD 3000?

We need $2000(1.03)^n > 3000 \Rightarrow (1.03)^n > 1.5$. Taking logarithms: $n\ln(1.03) > \ln(1.5) \Rightarrow n > \frac{\ln(1.5)}{\ln(1.03)} \approx \frac{0.405465}{0.029559} \approx 13.72$. The minimum number of complete years is 14. [M1 for setting up inequality, A1 for 14 years]

6 [Foundational] • Target: Fractions and division

Port B is 40 km from Port A on a bearing of 120°. Port C is 50 km from Port A on a bearing of 030°. Calculate the distance between Port B and Port C.

The angle between the two paths at Port A is $120^\circ - 030^\circ = 90^\circ$. Since this is a right triangle, use Pythagoras: $BC = \sqrt{40^2 + 50^2} = \sqrt{1600 + 2500} = \sqrt{4100} \approx 64.03$ km ≈ 64.0 km. [M1 for angle at A = 90, M1 for Pythagoras, A1 for 64.0]

7 [Foundational] • Target: Name, identify, order, and compare fractions

A drawer contains 5 blue socks, 4 grey socks, and 3 black socks. Two socks are pulled at random without replacement. What is the probability of pulling two socks of different colours?

Total socks = 12. Total pairs = $\binom{12}{2} = 66$. Number of same-colour pairs: $\binom{5}{2} + \binom{4}{2} + \binom{3}{2} = 10 + 6 + 3 = 19$. Therefore, the probability that the two socks have different colours is: $P(\text{Different}) = 1 - \frac{19}{66} = \frac{47}{66}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Equivalent and simplified fractions

Convert $0.1\dot{4}\dot{8}$ to a fraction in its simplest form.

Let $x = 0.1484848\ldots$. Then $10x = 1.484848\ldots$ and $1000x = 148.484848\ldots$. Subtract: $1000x - 10x = 148 - 1 = 147 \Rightarrow 990x = 147 \Rightarrow x = \frac{147}{990}$. Divide numerator and denominator by 3: $\frac{49}{330}$. [M1 for 1000x - 10x, A1 for 49/330]

9 [Intermediate] • Target: Adding and subtracting fractions

In a circle of radius 10 cm, an arc subtends a central angle of 72°. Calculate the perimeter of this sector to 1 decimal place.

Arc length: $s = \frac{72}{360} \times 2\pi(10) = \frac{1}{5} \times 20\pi = 4\pi \approx 12.57$ cm. Perimeter of sector = arc + 2 radii = $12.57 + 2(10) = 32.57 \approx 32.6$ cm. [M1 for arc length, M1 for adding 2r, A1 for 32.6]

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10 [Intermediate] • Target: Fractions of a quantity

A vertical pole of height 12 m casts a shadow of length 9 m on horizontal ground. Calculate the angle of elevation of the sun to the nearest degree.

Use tangent: $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{12}{9} \approx 1.333 \Rightarrow \theta = \tan^{-1}(1.333) \approx 53.1^\circ \approx 53^\circ$. [M1 for tan(theta)=12/9, A1 for 53]

11 [Intermediate] • Target: Adding and subtracting fractions

For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(5)$.

Final Answer: 33. Substitute $x = 5$: $f(5) = 6(5) + 3 = 33$.

12 [Intermediate] • Target: Adding and subtracting fractions

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

Final Answer: $x = 4$ and $x = 8$. Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

13 [Intermediate] • Target: Fractions of a quantity

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(2)$.

Final Answer: 8. Substitute $x = 2$: $f(2) = 2(2) + 4 = 8$.

14 [Intermediate] • Target: Fractions of a quantity

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Final Answer: $x = 1$ and $x = 6$. Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 8.01 Probability & Statistics

A data set recorded for fractions of a quantity consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

16 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 9.01 Functions

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(3)$.

Final Answer: 10. Substitute $x = 3$: $f(3) = 2(3) + 4 = 10$.

17 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 1.01 Number

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

Final Answer: $x = 2$ and $x = 7$. Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

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18 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 10.01 Sets & Venn Diagrams

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(4)$.

Final Answer: 12. Substitute $x = 4$: $f(4) = 2(4) + 4 = 12$.

19 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 3.01 Coordinate Geometry

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

Final Answer: $x = 3$ and $x = 8$. Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

20 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 4.01 Mensuration

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(5)$.

Final Answer: 14. Substitute $x = 5$: $f(5) = 2(5) + 4 = 14$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Name, identify, order, and compare fractions

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(2)$.

a. 16 ✓

b. 20

c. 8

d. 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

22 [Foundational] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

a. $x = 2$ and $x = 5$

b. $x = 2$ **and** $x = 4$ ✓

c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

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23 [Foundational] • Target: Name, identify, order, and compare fractions

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(5)$.

a. 28 ✓

b. 32

c. 20

d. 32

Substitute $x = 5$: $f(5) = 4(5) + 8 = 28$.

24 [Foundational] • Target: Equivalent and simplified fractions

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(2)$.

a. 19 ✓

b. 24

c. 10

d. 23

Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

25 [Foundational] • Target: Equivalent and simplified fractions

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

26 [Foundational] • Target: Equivalent and simplified fractions

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(5)$.

a. 34 ✓

b. 39

c. 25

d. 38

Substitute $x = 5$: $f(5) = 5(5) + 9 = 34$.

27 [Foundational] • Target: Adding and subtracting fractions

For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(2)$.

a. 15 ✓

b. 21

c. 12

d. 19

Substitute $x = 2$: $f(2) = 6(2) + 3 = 15$.

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28 [Foundational] • Target: Adding and subtracting fractions

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ **and** $x = 3$ ✓

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

30 [Intermediate] • Target: Name, identify, order, and compare fractions

A data set recorded for name, identify, order, and compare fractions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

31 [Intermediate] • Target: Name, identify, order, and compare fractions

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(3)$.

a. 20 ✓

b. 24

c. 12

d. 24

Substitute $x = 3$: $f(3) = 4(3) + 8 = 20$.

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32 [Intermediate] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ **and** $x = 5$ ✓

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

33 [Intermediate] • Target: Name, identify, order, and compare fractions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ **and** $x = 6$ ✓

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

34 [Intermediate] • Target: Equivalent and simplified fractions

A data set recorded for equivalent and simplified fractions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

35 [Intermediate] • Target: Equivalent and simplified fractions

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(3)$.

a. 24 ✓

b. 29

c. 15

d. 28

Substitute $x = 3$: $f(3) = 5(3) + 9 = 24$.

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36 [Intermediate] • Target: Equivalent and simplified fractions

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

37 [Intermediate] • Target: Equivalent and simplified fractions

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Adding and subtracting fractions • Cross-Lesson: 10.01 Sets & Venn Diagrams

A data set recorded for adding and subtracting fractions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

39 [Challenging] • Target: Adding and subtracting fractions • Cross-Lesson: 3.01 Coordinate Geometry

For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(3)$.

a. 21 ✓

b. 27

c. 18

d. 25

Substitute $x = 3$: $f(3) = 6(3) + 3 = 21$.

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40 [Challenging] • Target: Adding and subtracting fractions • Cross-Lesson: 4.01 Mensuration

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

41 [Challenging] • Target: Fractions of a quantity • Cross-Lesson: 5.01 Geometry

A data set recorded for fractions of a quantity consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

42 [Challenging] • Target: Multiplying fractions • Cross-Lesson: 6.01 Trigonometry

A data set recorded for multiplying fractions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

43 [Challenging] • Target: Multiplying fractions • Cross-Lesson: 7.01 Transformations & Vectors

For the model $f(x) = 3x + 5$ in multiplying fractions, evaluate $f(3)$.

a. 14 ✓

b. 17

c. 9

d. 18

Substitute $x = 3$: $f(3) = 3(3) + 5 = 14$.

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44 [Challenging] • Target: Fractions and division • Cross-Lesson: 8.01 Probability & Statistics

A data set recorded for fractions and division consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

45 [Challenging] • Target: Fractions and division • Cross-Lesson: 9.01 Functions

For the model $f(x) = 4x + 6$ in fractions and division, evaluate $f(3)$.

a. 18 ✓

b. 22

c. 12

d. 22

Substitute $x = 3$: $f(3) = 4(3) + 6 = 18$.

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